

business problems. They are structured, semi-structured, and unstructured, and come from many sources.

Acquire: In this stage, AI-based solutions support processing of all types of data from a variety of sources. Context management provides the models with relevant information from enterprise data sources. Caching enables faster responses.

Process: Generative AI uses foundation models. These are trained on huge quantities of unstructured and unlabelled data to perform specific tasks and act like a platform for other models. To process large amounts of unstructured text, foundation models leverage large language models (LLMs). Foundation models are fine tuned for domain adoption and to perform specific tasks better using short periods of training on labelled data. The process of further training a pre-trained model on a specific task or dataset to adapt it for a particular application or domain is called fine-tuning.

Store: Data is stored in this stage.

Train: This stage primarily consists of model hub, blob storage, databases, and a prompt library. Model hub consists of trained and approved models that can be provisioned on demand. It acts as a repository for model checkpoints, weights, and parameters.

Prompt engineering is a process of designing, refining, and optimising input prompts to guide a generative AI model towards producing the desired outputs.

Deliver: This stage covers cognitive search. The model provides access to the right data at the right time to produce accurate output. Stakeholders use API gateway channels to interact with enterprises. This is a single point of entry for consumers to access back-end services.

AI security: This helps in establishing strong security. AI security must cover strategy, planning and intellectual property. The generative

AI platform needs to provide robust encryption, multi-level user access, authentication, and authorisation.

Model consumers: Various stakeholders, both internal and external, are part of this layer. They are the primary users of the systems.

Use cases of generative AI

Healthcare and pharma

Generative AI based applications help healthcare professionals be more productive, identifying potential issues upfront, providing insights to deliver interconnected health and improve patient outcomes. It helps in:

- Processing claims, scheduling appointments, and managing medical records
- Writing patient health summaries
- Accelerating the speed and quality of care. It can also improve drug adherence
- Developing individualised treatment plans based on a patient's genetic makeup, medical history, lifestyle, etc
- Building healthcare virtual assistants

Manufacturing

Generative AI enables manufacturers to create more with their data, leading to advancements in predictive maintenance and demand forecasting. It also helps in simulating manufacturing quality, improving production speed, and materials efficiency.

Retail

Generative AI helps in personalising offerings, brand management, and optimising marketing and sales activities.

- *Personalised offerings:* Enables retailers to deliver customised experiences, offerings, pricing, and planning. It also helps in modernising the online and physical buying experience.
- *Dynamic pricing and planning:* It helps predict demand for different products, providing greater confidence for pricing and stocking decisions.

Banking

Generative AI applications help in delivering a personalised banking experience to customers. It enables risk analytics and fraud prevention, risk mitigation and portfolio optimisation, as well as customer pattern analysis and customer financial planning.

Insurance

GenAI's ability to analyse and process large amounts of data helps in accurate risk assessments and effective claims processes. Various data categories are customer feedback, claims records, policy records, etc.

Education

Generative AI helps to connect teachers and students. It also enables collaboration between teachers, administrators and technology innovators to help educational institutions provide better education.

Telecommunications

Generative AI adoption by the telecom industry improves operations efficiency and network performance. In this industry, generative AI can be used to:

- Analyse customer purchasing patterns
- Offer personalised recommendations of services
- Enhance sales
- Manage customer loyalty
- Give insights into customer preferences
- Improve data and network security, enhancing fraud detection

Public sector

The goal of digital governments is to establish a connected government and provide better citizen services. Generative AI enables these citizen services to be delivered more effectively while protecting confidential information.

- *Smart cities:* Generative AI helps in toll management, traffic optimisation, and sustainability.